

Entropic Optimal Transport and Sinkhorn

Statements, hints and worked solutions

Each exercise is followed by question-by-question hints and a worked solution. Try the questions before reading the hints.

Exercise 1 (2x2 entropic OT with perturbation).

Let

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad a = \begin{pmatrix} p \\ 1-p \end{pmatrix}, \quad b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix},$$

with $p \in (0, 1)$, and let $\varepsilon > 0$. Consider

$$\min_{P \in \mathbb{R}_+^{2 \times 2}} \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} (\log P_{i,j} - 1) \quad \text{subject to} \quad P\mathbf{1} = a, \quad P^\top \mathbf{1} = b.$$

(1) Show that every admissible transport plan can be written as

$$P(x) = \begin{pmatrix} x & y(x) \\ z(x) & w(x) \end{pmatrix}$$

for a single scalar x , and give explicit formulas for $y(x)$, $z(x)$, $w(x)$, as well as the admissible interval $I(p)$ for x .

(2) Rewrite the objective as a scalar function $F_\varepsilon(x)$ and compute $F'_\varepsilon(x)$.

(3) Explain why the optimal $x_\varepsilon(p)$ is necessarily in the interior of $I(p)$ for $0 < p < 1$. Optional: from the optimality condition $F'_\varepsilon(x) = 0$, give the expression of the optimal $x_\varepsilon(p)$.

(4) Optional: assume $0 < p < 1/2$. As $\varepsilon \downarrow 0$, obtain an expansion $x_\varepsilon(p) = p + c(p)e^{-2/\varepsilon} + o(e^{-2/\varepsilon})$ and determine $c(p)$.

Useful hints

(1) Set $x = P_{1,1}$ and eliminate the other entries using the marginal constraints.

(2) Use $[r(\log r - 1)]' = \log r$.

(3) Inspect the signs of the derivative at the two endpoints, and its strict monotonicity.

(4) Put $x = p - \delta$ in the optimality equation and balance the leading terms as $e^{-2/\varepsilon} \rightarrow 0$.

Detailed solution

(1) Let

$$P = \begin{pmatrix} x & y \\ z & w \end{pmatrix}.$$

The constraints $P\mathbf{1} = a$ and $P^\top \mathbf{1} = b$ are

$$x + y = p, \quad z + w = 1 - p, \quad x + z = \frac{1}{2}, \quad y + w = \frac{1}{2}.$$

From $x + y = p$ we get $y = p - x$. From $x + z = \frac{1}{2}$ we get $z = \frac{1}{2} - x$. Then w is fixed by $z + w = 1 - p$:

$$w = 1 - p - z = 1 - p - \left(\frac{1}{2} - x\right) = \frac{1}{2} - p + x.$$

Hence every feasible plan is

$$P(x) = \begin{pmatrix} x & p - x \\ \frac{1}{2} - x & \frac{1}{2} - p + x \end{pmatrix}.$$

Nonnegativity gives

$$x \geq 0, \quad p - x \geq 0, \quad \frac{1}{2} - x \geq 0, \quad \frac{1}{2} - p + x \geq 0.$$

Thus

$$x \in \left[\max(0, p - \frac{1}{2}), \min(p, \frac{1}{2}) \right] =: I_p.$$

(2) The cost part is

$$\langle C, P(x) \rangle = (p - x) + \left(\frac{1}{2} - x \right) = p + \frac{1}{2} - 2x.$$

For the entropic part set

$$H(x) := x(\log x - 1) + (p - x)(\log(p - x) - 1) + \left(\frac{1}{2} - x\right)(\log(\frac{1}{2} - x) - 1) + \left(\frac{1}{2} - p + x\right)(\log(\frac{1}{2} - p + x) - 1).$$

Then

$$F_\varepsilon(x) = p + \frac{1}{2} - 2x + \varepsilon H(x), \quad x \in I_p.$$

Differentiate term by term. Using

$$\frac{d}{dx} [x(\log x - 1)] = \log x, \quad \frac{d}{dx} [(p - x)(\log(p - x) - 1)] = -\log(p - x),$$

$$\frac{d}{dx} \left[\left(\frac{1}{2} - x\right)(\log(\frac{1}{2} - x) - 1) \right] = -\log(\frac{1}{2} - x), \quad \frac{d}{dx} \left[\left(\frac{1}{2} - p + x\right)(\log(\frac{1}{2} - p + x) - 1) \right] = \log(\frac{1}{2} - p + x),$$

we get

$$F'_\varepsilon(x) = -2 + \varepsilon \left[\log x - \log(p - x) - \log(\frac{1}{2} - x) + \log(\frac{1}{2} - p + x) \right].$$

(3) In the interior, $F''_\varepsilon = \varepsilon(1/x + 1/(p - x) + 1/(1/2 - x) + 1/(1/2 - p + x)) > 0$. At the left endpoint $F'_\varepsilon \rightarrow -\infty$; at the right endpoint $F'_\varepsilon \rightarrow +\infty$. Thus there is a unique interior minimizer. The condition $F'_\varepsilon(x) = 0$ is

$$-2 + \varepsilon \log \left(\frac{x(\frac{1}{2} - p + x)}{(p - x)(\frac{1}{2} - x)} \right) = 0,$$

that is,

$$\frac{x(\frac{1}{2} - p + x)}{(p - x)(\frac{1}{2} - x)} = \exp\left(\frac{2}{\varepsilon}\right) =: \tau.$$

Equivalently,

$$x(\frac{1}{2} - p + x) = \tau(p - x)(\frac{1}{2} - x), \quad \tau = e^{2/\varepsilon}.$$

This is a quadratic equation in x . Expanding and collecting terms one gets

$$(1 - \tau)x^2 + \left(p\tau - p + \frac{\tau}{2} + \frac{1}{2}\right)x - \frac{p\tau}{2} = 0.$$

Set

$$a = 1 - \tau, \quad b = p\tau - p + \frac{\tau + 1}{2}, \quad c = -\frac{p\tau}{2}.$$

Then x solves

$$ax^2 + bx + c = 0.$$

Thus

$$x_\varepsilon(p) = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{b^2 - 2p\tau(\tau - 1)}}{2(1 - \tau)}.$$

Among the two roots, the one in $I_p = [\max(0, p - \frac{1}{2}), \min(p, \frac{1}{2})]$ is the minimizer:

$$x_\varepsilon(p) = \frac{-b + \sqrt{b^2 - 2p\tau(\tau - 1)}}{2(1 - \tau)}, \quad b = p\tau - p + \frac{\tau + 1}{2}, \quad \tau = e^{2/\varepsilon}.$$

(4) Asymptotics as $\varepsilon \rightarrow 0^+$ (i.e. $\tau \rightarrow +\infty$). Assume $p < \frac{1}{2}$. Then the unregularized solution is $x_0(p) = p$ (right endpoint of I_p). For $\varepsilon > 0$ small we look for

$$x_\varepsilon = p - \delta_\varepsilon, \quad \delta_\varepsilon > 0, \quad \delta_\varepsilon \ll 1.$$

Plug into the optimality equation:

$$\frac{(p - \delta_\varepsilon)(\frac{1}{2} - p + p - \delta_\varepsilon)}{(p - (p - \delta_\varepsilon))(\frac{1}{2} - (p - \delta_\varepsilon))} = \tau.$$

This is

$$\frac{(p - \delta_\varepsilon)(\frac{1}{2} - \delta_\varepsilon)}{\delta_\varepsilon(\frac{1}{2} - p + \delta_\varepsilon)} = \tau.$$

For small δ_ε ,

$$(p - \delta_\varepsilon)(\frac{1}{2} - \delta_\varepsilon) = \frac{p}{2} + O(\delta_\varepsilon), \quad \frac{1}{2} - p + \delta_\varepsilon = \frac{1}{2} - p + O(\delta_\varepsilon).$$

Hence, at leading order,

$$\frac{\frac{p}{2}}{\delta_\varepsilon(\frac{1}{2} - p)} \simeq \tau \implies \delta_\varepsilon \simeq \frac{p}{2(\frac{1}{2} - p)} \tau^{-1} = \frac{p}{1 - 2p} e^{-2/\varepsilon}.$$

Therefore

$$x_\varepsilon(p) = p - \delta_\varepsilon = p - \frac{p}{1 - 2p} \tau^{-1} + o(\tau^{-1}) = p - \frac{p}{1 - 2p} e^{-2/\varepsilon} + o(e^{-2/\varepsilon}).$$

For $p > \frac{1}{2}$ the same argument, exchanging the two rows, gives the symmetric boundary layer near $x = \frac{1}{2}$.

Exercise 2 (Dual of entropic-regularized linear program).

Consider the entropic-regularized linear program

$$\min_{P \geq 0, A(P)=b} \langle C, P \rangle + \varepsilon H(P), \quad H(P) := \sum_i P_i (\log P_i - 1),$$

where P is a finite-dimensional nonnegative vector (think of $P \in \mathbb{R}_+^N$, or a matrix reshaped as a vector), $C \in \mathbb{R}^N$, $A : \mathbb{R}^N \rightarrow \mathbb{R}^m$ is linear, $b \in \mathbb{R}^m$, and $\varepsilon > 0$. Assume strict feasibility: there exists $P^0 > 0$ with $A(P^0) = b$. Use $0 \log 0 = 0$.

- (1) Write the Lagrangian $\mathcal{L}(P, \lambda)$.
- (2) Assuming we can swap $\inf_P$ and $\sup_{\lambda}$, write the dual problem.
- (3) Give the primal–dual optimality relation between P and λ .
- (4) Give the expression of $A(P)$ in the case of the marginal constraint of classical OT. Compute $A^\top$, and show that the dual you obtained matches the classical dual of entropic OT.

Useful hints

- (1) Use the sign convention $\langle \lambda, b - A(P) \rangle$.
- (2) Minimize $dp + \varepsilon p(\log p - 1)$ over $p \geq 0$, entry by entry.
- (3) Combine scalar stationarity with feasibility.
- (4) Compute $\langle (f, g), A(P) \rangle$ and collect the coefficient of $P_{i,j}$.

Detailed solution

(1) Choose

$$\mathcal{L}(P, \lambda) = \langle C, P \rangle + \varepsilon H(P) + \langle \lambda, b - A(P) \rangle.$$

The scalar function $dp + \varepsilon p(\log p - 1)$ has its unique minimum at $p = e^{-d/\varepsilon}$, with minimum value $-\varepsilon e^{-d/\varepsilon}$.

(2) Applying this to $d = C_i - (A^\top \lambda)_i$ gives

$$\sup_{\lambda \in \mathbb{R}^m} \left\{ \langle b, \lambda \rangle - \varepsilon \sum_i e^{((A^\top \lambda)_i - C_i)/\varepsilon} \right\}.$$

The primal objective is coercive because $H(P)$ grows faster than linearly. It has a unique minimizer on the nonempty closed feasible set. Strict feasibility ensures strong duality and a finite optimal multiplier; it also excludes forced zero coordinates.

(3) The optimality equations are

$$P_i^\star = e^{((A^\top \lambda^\star)_i - C_i)/\varepsilon}, \quad A(P^\star) = b.$$

Conversely, these equations certify optimality by equality in the scalar minimizations. Multipliers need not be unique if the constraints are redundant.

(4) For an $n \times m$ coupling matrix, write the constraint vector as (a, b) instead of the generic b . Then

$$A(P) = (P \mathbf{1}_m, P^\top \mathbf{1}_n), \quad (A^\top(f, g))_{i,j} = f_i + g_j.$$

Indeed, $\langle f, P \mathbf{1}_m \rangle + \langle g, P^\top \mathbf{1}_n \rangle = \sum_{i,j} (f_i + g_j) P_{i,j}$. The dual is therefore

$$\sup_{f, g} \left\{ \langle a, f \rangle + \langle b, g \rangle - \varepsilon \sum_{i,j} e^{(f_i + g_j - C_{i,j})/\varepsilon} \right\}.$$

With $u_i = e^{f_i/\varepsilon}$, $v_j = e^{g_j/\varepsilon}$, and $K_{i,j} = e^{-C_{i,j}/\varepsilon}$, the primal relation is $P_{i,j}^\star = u_i K_{i,j} v_j$. Positive equal-mass marginals supply strict feasibility.

Exercise 3 (Entropic multimarginal OT with 3 marginals).

Let $a \in \mathbb{R}_{++}^{n_1}$, $b \in \mathbb{R}_{++}^{n_2}$, $c \in \mathbb{R}_{++}^{n_3}$ of equal total mass, and let $\varepsilon > 0$ and $C \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. For $T \in \mathbb{R}_+^{n_1 \times n_2 \times n_3}$ define the three marginalization operators

$$\pi_1(T) := \left(\sum_{j,k} T_{i,j,k} \right)_{i=1}^{n_1}, \quad \pi_2(T) := \left(\sum_{i,k} T_{i,j,k} \right)_{j=1}^{n_2}, \quad \pi_3(T) := \left(\sum_{i,j} T_{i,j,k} \right)_{k=1}^{n_3}.$$

Consider the problem

$$\min_{T \geq 0} \langle C, T \rangle + \varepsilon \sum_{i,j,k} T_{i,j,k} (\log T_{i,j,k} - 1) \quad \text{s.t.} \quad \pi_1(T) = a, \pi_2(T) = b, \pi_3(T) = c.$$

- (1) Write the Lagrangian using multipliers f, g, h for π_1, π_2, π_3 .
- (2) Assuming one can swap $\min_T$ and $\max_{f,g,h}$, write the dual.
- (3) Write the Sinkhorn iterations in the log domain for the three potentials.
- (4) (Bonus) Explain why the usual Hilbert-metric/Birkhoff argument for 2 marginals does not give a contraction here.

Useful hints

- (1) Use the constraints with signs $\langle f, a - \pi_1(T) \rangle$, and similarly for the other two.
- (2) Apply the scalar entropy conjugate to each tensor entry.
- (3) Enforce one marginal at a time, using the most recently updated potentials.
- (4) Changing two input scalings contributes two projective errors, rather than a single linear-map error.

Detailed solution

- (1) Introduce $f \in \mathbb{R}^{n_1}, g \in \mathbb{R}^{n_2}, h \in \mathbb{R}^{n_3}$. Using the shorthand

$$\langle f, \pi_1(T) \rangle = \sum_i f_i \sum_{j,k} T_{i,j,k}, \quad \text{etc.},$$

the Lagrangian is

$$\mathcal{L}(T, f, g, h) = \langle C, T \rangle + \varepsilon \sum_{i,j,k} T_{i,j,k} (\log T_{i,j,k} - 1) + \langle f, a - \pi_1(T) \rangle + \langle g, b - \pi_2(T) \rangle + \langle h, c - \pi_3(T) \rangle.$$

Grouping terms in T :

$$\mathcal{L}(T, f, g, h) = \langle a, f \rangle + \langle b, g \rangle + \langle c, h \rangle + \sum_{i,j,k} (C_{i,j,k} - f_i - g_j - h_k) T_{i,j,k} + \varepsilon \sum_{i,j,k} T_{i,j,k} (\log T_{i,j,k} - 1).$$

- (2) For fixed (f, g, h) we minimize over $T \geq 0$ termwise. For one entry $t \geq 0$ with coefficient $d := C_{i,j,k} - f_i - g_j - h_k$:

$$\min_{t \geq 0} d t + \varepsilon t (\log t - 1) \Rightarrow d + \varepsilon \log t = 0 \Rightarrow t = \exp\left(\frac{f_i + g_j + h_k - C_{i,j,k}}{\varepsilon}\right).$$

Value at the minimum:

$$-\varepsilon \exp\left(\frac{f_i + g_j + h_k - C_{i,j,k}}{\varepsilon}\right).$$

Hence the dual function is

$$g(f, g, h) = \langle a, f \rangle + \langle b, g \rangle + \langle c, h \rangle - \varepsilon \sum_{i,j,k} \exp\left(\frac{f_i + g_j + h_k - C_{i,j,k}}{\varepsilon}\right).$$

So the dual problem is

$$\max_{f,g,h} \langle a, f \rangle + \langle b, g \rangle + \langle c, h \rangle - \varepsilon \sum_{i,j,k} \exp\left(\frac{f_i + g_j + h_k - C_{i,j,k}}{\varepsilon}\right).$$

- (3) Let $K_{i,j,k} := e^{-C_{i,j,k}/\varepsilon}$ and define scaling vectors

$$u_i := e^{f_i/\varepsilon}, \quad v_j := e^{g_j/\varepsilon}, \quad w_k := e^{h_k/\varepsilon}.$$

The primal optimal tensor is then

$$T_{i,j,k} = K_{i,j,k} u_i v_j w_k.$$

Imposing the three marginal constraints compactly:

$$\pi_1(K \odot (u \otimes v \otimes w)) = a, \quad \pi_2(K \odot (u \otimes v \otimes w)) = b, \quad \pi_3(K \odot (u \otimes v \otimes w)) = c,$$

where $\odot$ is entrywise product and $(u \otimes v \otimes w)_{i,j,k} = u_i v_j w_k$. This yields the 3-block Sinkhorn updates:

$$\begin{aligned} u_i^{(\ell+1)} &= \frac{a_i}{[\pi_1(K \odot (1 \otimes v^{(\ell)} \otimes w^{(\ell)}))]_i} = \frac{a_i}{\sum_{j,k} K_{i,j,k} v_j^{(\ell)} w_k^{(\ell)}}, \\ v_j^{(\ell+1)} &= \frac{b_j}{\sum_{i,k} K_{i,j,k} u_i^{(\ell+1)} w_k^{(\ell)}}, \\ w_k^{(\ell+1)} &= \frac{c_k}{\sum_{i,j} K_{i,j,k} u_i^{(\ell+1)} v_j^{(\ell+1)}}. \end{aligned}$$

In log form ($f_i = \varepsilon \log u_i$, etc.):

$$\begin{aligned} f_i^{(\ell+1)} &= \varepsilon \log a_i - \varepsilon \log \left(\sum_{j,k} K_{i,j,k} e^{g_j^{(\ell)}/\varepsilon} e^{h_k^{(\ell)}/\varepsilon} \right), \\ g_j^{(\ell+1)} &= \varepsilon \log b_j - \varepsilon \log \left(\sum_{i,k} K_{i,j,k} e^{f_i^{(\ell+1)}/\varepsilon} e^{h_k^{(\ell)}/\varepsilon} \right), \\ h_k^{(\ell+1)} &= \varepsilon \log c_k - \varepsilon \log \left(\sum_{i,j} K_{i,j,k} e^{f_i^{(\ell+1)}/\varepsilon} e^{g_j^{(\ell+1)}/\varepsilon} \right). \end{aligned}$$

(4) Bonus. Each two-marginal half-step is a positive linear map followed by componentwise inversion. With three marginals, the denominator for u depends on $v \otimes w$, and

$$d_H(v \otimes w, \tilde{v} \otimes \tilde{w}) = d_H(v, \tilde{v}) + d_H(w, \tilde{w}).$$

Viewing tensor contraction as a positive linear map on this tensor-product input gives a bound of the form $d_H(u^{\text{new}}, \tilde{u}^{\text{new}}) \leq \lambda [d_H(v, \tilde{v}) + d_H(w, \tilde{w})]$, where $\lambda < 1$. This is not automatically a contraction in the maximum of the block errors, since its bound is 2λ . It does not prove failure of convergence; it explains why the two-block proof cannot simply be reused. A Birkhoff coefficient itself never exceeds 1.

Exercise 4 (Gaussian example for Sinkhorn in dual form).

Let α and β be probability measures on $\mathbb{R}$ and let $c(x, y) = \frac{1}{2}(x - y)^2$. The entropic OT dual can be written

$$\max_{f,g} \int f(x) d\alpha(x) + \int g(y) d\beta(y) - \varepsilon \int \int \exp\left(\frac{f(x)+g(y)-c(x,y)}{\varepsilon}\right) d\alpha(x) d\beta(y).$$

(1) Recall the Sinkhorn update on f (dual form).

(2) Assume $\alpha = \mathcal{N}(m_1, \sigma_1^2)$, $\beta = \mathcal{N}(m_2, \sigma_2^2)$, with $\sigma_1, \sigma_2 > 0$, and let $g(y) = \frac{1}{2}ay^2 + by + c_0$, where $a < 1 + \varepsilon/\sigma_2^2$ (the integrability condition). Compute the integral on the right-hand side and show that the updated f is also quadratic.

(3) Give an intuitive explanation of why (for this quadratic/Gaussian setting) the optimal dual potentials remain quadratic, and what this implies for the optimal coupling $\pi^\star$.

Useful hints

- (1) Maximize the dual in $f(x)$ pointwise.
- (2) Complete the square in y and retain the Gaussian normalization factor.
- (3) Try a Gaussian joint law with cross-covariance q and match its cross precision with that of the Gibbs kernel.

Detailed solution

(1) Pointwise stationarity in f gives

$$f^{\text{new}}(x) = -\varepsilon \log \int e^{(g(y) - (x-y)^2/2)/\varepsilon} d\beta(y).$$

The displayed dual in the statement omits the constant $+\varepsilon$ of the probability-normalized KL convention; this changes neither potentials nor optimizer.

(2) Collect the exponent as $-Ay^2/2 + B(x)y + D(x)$, where

$$A = \frac{1-a}{\varepsilon} + \frac{1}{\sigma_2^2} > 0, \quad B(x) = \frac{x+b}{\varepsilon} + \frac{m_2}{\sigma_2^2}, \quad D(x) = \frac{c_0 - x^2/2}{\varepsilon} - \frac{m_2^2}{2\sigma_2^2}.$$

Completing the square yields

$$\int e^{(g(y) - (x-y)^2/2)/\varepsilon} d\beta(y) = \frac{e^{D(x) + B(x)^2/(2A)}}{\sigma_2 \sqrt{A}}.$$

Thus $f^{\text{new}}(x) = \varepsilon \log(\sigma_2 \sqrt{A}) - \varepsilon D(x) - \varepsilon B(x)^2/(2A)$ is quadratic. Without $A > 0$, the integral need not be finite, so the qualification in the statement is essential.

(3) Quadratic updates suggest Gaussian closure, but invariance of an iteration family alone is not an existence proof. To verify the optimizer directly, take a Gaussian coupling with prescribed means and covariance

$$\begin{pmatrix} \sigma_1^2 & q \\ q & \sigma_2^2 \end{pmatrix}, \quad q = \frac{\sqrt{\varepsilon^2 + 4\sigma_1^2\sigma_2^2} - \varepsilon}{2}.$$

Its determinant is $\varepsilon q > 0$. Therefore its off-diagonal precision entry is $-q/(\varepsilon q) = -1/\varepsilon$. In its log density ratio relative to $\alpha \otimes \beta$, the mixed term is xy/ε , exactly as in $-(x-y)^2/(2\varepsilon)$. The remaining terms are separate quadratics in x, y , giving potentials f, g and the exact relation $d\pi^*/d(\alpha \otimes \beta) = e^{(f+g-c)/\varepsilon}$. Together with the prescribed marginals, this certifies primal–dual optimality. Strict convexity of entropy makes the coupling unique.

Exercise 5 (Envelope theorem and gradients of the entropic cost).

Recall the envelope theorem in the finite-dimensional smooth case:

Let $V(\theta) := \min_z F(z, \theta)$ with F smooth, with the minimization over a fixed compact set, and for each θ assume there is a unique minimizer $z^*(\theta)$. Then

$$\nabla_\theta V(\theta) = \nabla_\theta F(z^*(\theta), \theta),$$

i.e. the derivative of V w.r.t. θ is the partial derivative of F at the optimizer (no derivative of z^* needed).

Let $a \in \mathbb{R}_{++}^m, b \in \mathbb{R}_{++}^n$ be probability histograms, let $\varepsilon > 0$, and $C \in \mathbb{R}^{m \times n}$ a cost matrix. The entropic OT cost is

$$\mathcal{W}_\varepsilon(a, b; C) := \min_{\substack{P \in \mathbb{R}_{++}^{m \times n} \\ P\mathbf{1} = a, P^\top \mathbf{1} = b}} \left\{ \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} (\log P_{i,j} - 1) \right\}.$$

- (1) Using the envelope theorem on this *primal* problem, give the gradient $\nabla_C \mathcal{W}_\varepsilon(a, b; C)$.
- (2) Assume the cost comes from positions: $x = (x_1, \dots, x_m) \in \mathbb{R}^d$, $y = (y_1, \dots, y_n) \in \mathbb{R}^d$, and

$$C_{i,j} = \frac{1}{2} \|x_i - y_j\|^2.$$

Compute $\nabla_{x_1} \mathcal{W}_\varepsilon(a, b; C)$.

- (3) Recall the dual of entropic OT

$$\mathcal{W}_\varepsilon(a, b; C) = \max_{f \in \mathbb{R}^m, g \in \mathbb{R}^n} \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right).$$

Using again the envelope theorem, compute its derivative along perturbations $\dot{a}$ satisfying $\sum_i \dot{a}_i = 0$, and explain the additive-constant ambiguity.

Useful hints

- (1) Differentiate the objective in C , holding the optimal plan fixed.
- (2) Only the first row of costs depends on x_1 .
- (3) Work on the probability simplex. Potentials differing by a constant give the same pairing with a zero-sum perturbation.

Detailed solution

- (1) Write

$$F(P, C) = \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} (\log P_{i,j} - 1), \quad \mathcal{W}_\varepsilon(a, b; C) = \min_{P \in \Pi(a,b)} F(P, C),$$

where $\Pi(a, b)$ is the transport polytope. For fixed (a, b, C) , let $P_\varepsilon^\star(a, b; C)$ be the unique optimal coupling. By the envelope theorem,

$$\nabla_C \mathcal{W}_\varepsilon(a, b; C) = \nabla_C F(P_\varepsilon^\star, C) = P_\varepsilon^\star.$$

Since C and P have the same shape, this means

$$\frac{\partial \mathcal{W}_\varepsilon}{\partial C_{i,j}} = (P_\varepsilon^\star)_{i,j}.$$

- (2) Now $C_{1,j} = \frac{1}{2} \|x_1 - y_j\|^2$. By the chain rule,

$$\nabla_{x_1} \mathcal{W}_\varepsilon(a, b; C) = \sum_{j=1}^n \frac{\partial \mathcal{W}_\varepsilon}{\partial C_{1,j}} \nabla_{x_1} C_{1,j} = \sum_{j=1}^n (P_\varepsilon^\star)_{1,j} (x_1 - y_j).$$

Hence

$$\nabla_{x_1} \mathcal{W}_\varepsilon(a, b; C) = x_1 \sum_j (P_\varepsilon^\star)_{1,j} - \sum_j (P_\varepsilon^\star)_{1,j} y_j.$$

But $\sum_j (P_\varepsilon^\star)_{1,j} = a_1$, so

$$\nabla_{x_1} \mathcal{W}_\varepsilon(a, b; C) = a_1 x_1 - \sum_j (P_\varepsilon^\star)_{1,j} y_j.$$

- (3) Consider the dual

$$D(f, g; a, b) = \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right).$$

Let (f^*, g^*) be the maximizers. By the envelope theorem (now on the dual side),

$$\nabla_a \mathcal{W}_\varepsilon(a, b; C) = f^*, \quad \nabla_b \mathcal{W}_\varepsilon(a, b; C) = g^*.$$

These equalities are understood on the tangent spaces of the probability simplices: $D_a \mathcal{W}_\varepsilon[\dot{a}] = \langle f^*, \dot{a} \rangle$ for $\sum_i \dot{a}_i = 0$, and similarly for b . The gauge $(f^*, g^*) \mapsto (f^* + c, g^* - c)$ does not change these derivatives. The orthogonal tangent-space representative is $f^* - \frac{1}{m}(\sum_i f_i^*)\mathbf{1}_m$, not an intrinsically defined ambient gradient.

Exercise 6 (Entropic OT, joint convexity, and semi-relaxed reformulation).

Let $a = (a_1, \dots, a_m) \in \Delta_m$ and $b = (b_1, \dots, b_n) \in \Delta_n$ be probability vectors and let $C \in \mathbb{R}^{m \times n}$ be a cost matrix. Consider the (balanced) entropic OT problem

$$W_\varepsilon(a, b) := \min_{\substack{P \in \mathbb{R}_+^{m \times n} \\ P\mathbf{1}_n = a, P^\top \mathbf{1}_m = b}} \left\{ \langle C, P \rangle + \varepsilon H(P) \right\}, \quad H(P) := \sum_{i,j} P_{i,j} (\log P_{i,j} - 1).$$

It is well known (and you may take it for granted) that the dual of this problem is

$$W_\varepsilon(a, b) = \sup_{f \in \mathbb{R}^m, g \in \mathbb{R}^n} \left\{ \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i=1}^m \sum_{j=1}^n \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right) \right\}.$$

(1) Using the dual expression above, show that *both* maps

$$a \mapsto W_\varepsilon(a, b) \quad \text{and} \quad b \mapsto W_\varepsilon(a, b)$$

are convex, and in fact that $W_\varepsilon(a, b)$ is jointly convex in (a, b) .

(2) Show that the outer problem

$$\min_{a \in \Delta_m} W_\varepsilon(a, b)$$

is equivalent to the semi-relaxed problem

$$\min_{\substack{P \geq 0 \\ P^\top \mathbf{1}_m = b}} \langle C, P \rangle + \varepsilon H(P).$$

(3) Solve this semi-relaxed problem in closed form.

Useful hints

- (1) A supremum of affine functions is convex. Check which entropy convention is being used.
- (2) The free source marginal is forced to be $P\mathbf{1}_n$.
- (3) Minimize independently in each column with a single mass constraint.

Detailed solution

(1) **Joint convexity.** From the dual

$$W_\varepsilon(a, b) = \sup_{f, g} (\langle f, a \rangle + \langle g, b \rangle - R(f, g)), \quad R(f, g) := \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right),$$

we see that for each fixed (f, g) the map $(a, b) \mapsto \langle f, a \rangle + \langle g, b \rangle - R(f, g)$ is affine in (a, b) (since R does not depend on (a, b)). A supremum of affine functions is convex, hence W_ε is convex in a , convex in b , and actually jointly convex in (a, b) .

(2) Semi-relaxed reformulation. Write

$$\min_{a \in \Delta_m} W_\varepsilon(a, b) = \min_{a \in \Delta_m} \min_{\substack{P \geq 0 \\ P \mathbf{1}_n = a \\ P^\top \mathbf{1}_m = b}} \{ \langle C, P \rangle + \varepsilon H(P) \}.$$

For any $P \geq 0$ with $P^\top \mathbf{1}_m = b$, choosing $a = P \mathbf{1}_n$ makes P feasible for the inner problem. Conversely, any feasible pair (P, a) satisfies $a = P \mathbf{1}_n$ anyway. Thus we can eliminate a and obtain the equivalent problem

$$\min_{\substack{P \geq 0 \\ P^\top \mathbf{1}_m = b}} \langle C, P \rangle + \varepsilon H(P).$$

(3) Closed form. The semi-relaxed problem separates over columns. For a fixed j , we solve

$$\min_{p_1, \dots, p_m \geq 0} \sum_{i=1}^m C_{i,j} p_i + \varepsilon \sum_{i=1}^m p_i (\log p_i - 1) \quad \text{s.t.} \quad \sum_{i=1}^m p_i = b_j.$$

Using a Lagrange multiplier λ_j for the column constraint, stationarity gives

$$C_{i,j} + \varepsilon \log p_i - \lambda_j = 0 \quad \implies \quad p_i = \exp\left(\frac{\lambda_j - C_{i,j}}{\varepsilon}\right).$$

Imposing $\sum_i p_i = b_j$ yields

$$e^{\lambda_j/\varepsilon} = \frac{b_j}{\sum_{k=1}^m e^{-C_{k,j}/\varepsilon}},$$

hence

$$p_i = \frac{b_j e^{-C_{i,j}/\varepsilon}}{\sum_{k=1}^m e^{-C_{k,j}/\varepsilon}}.$$

Doing this for each column gives $P_{i,j}^* = b_j e^{-C_{i,j}/\varepsilon} / \sum_k e^{-C_{k,j}/\varepsilon}$. If $b_j = 0$, the entire column is zero and the same formula still holds. This joint convexity concerns the entropy-only convention above; subtracting marginal entropies to use $\text{KL}(P \mid a \otimes b)$ changes that conclusion.

Exercise 7 (Invariance and rescaling (continuous setting)).

Let X, Y be compact metric spaces, α, β probability measures, and $c : X \times Y \rightarrow \mathbb{R}$ a continuous cost and $\varepsilon > 0$, consider the entropic optimal transport problem

$$\min_{\pi \in \Pi(\alpha, \beta)} \left\{ \int_{X \times Y} c(x, y) d\pi(x, y) + \varepsilon \int_{X \times Y} \left(\log \frac{d\pi}{d(\alpha \otimes \beta)}(x, y) - 1 \right) d\pi(x, y) \right\},$$

where $\Pi(\alpha, \beta)$ is the set of couplings of α and β and the entropy is defined as $+\infty$ if $\pi \not\ll \alpha \otimes \beta$.

(1) (Invariance under potentials) Let $u : X \rightarrow \mathbb{R}$ and $v : Y \rightarrow \mathbb{R}$ be integrable. Show that replacing the cost by

$$c'(x, y) = c(x, y) + u(x) + v(y)$$

does *not* change the optimal plan: the minimizer for c' is the same coupling as for c .

(2) (Scaling of the cost) For $\lambda > 0$, denote by $\pi_{c, \varepsilon}$ the optimal coupling for cost c and regularization ε , and by $\pi_{\lambda c, \varepsilon}$ the one for the scaled cost λc . Show that

$$\pi_{\lambda c, \varepsilon} = \pi_{c, \varepsilon/\lambda},$$

i.e. scaling the cost by λ is the same as keeping the cost and scaling the regularization by $1/\lambda$.

Useful hints

- (1) Marginal-only terms integrate to constants on the feasible set.
- (2) Divide the entire objective by $\lambda > 0$.

Detailed solution

- (1) For every feasible coupling,

$$\int (c + u + v) d\pi = \int c d\pi + \int u d\alpha + \int v d\beta.$$

The last two integrals are finite constants independent of π , while the entropy and feasible set are unchanged. Therefore the minimizer is unchanged. The hypotheses give a finite feasible competitor $\alpha \otimes \beta$ and a unique entropic optimizer.

- (2) Writing $H_{\alpha \otimes \beta}(\pi)$ for the entropy appearing in the statement,

$$\int \lambda c d\pi + \varepsilon H_{\alpha \otimes \beta}(\pi) = \lambda \left(\int c d\pi + \frac{\varepsilon}{\lambda} H_{\alpha \otimes \beta}(\pi) \right).$$

Multiplication by a positive constant does not change the argmin. Thus $\pi_{\lambda c, \varepsilon} = \pi_{c, \varepsilon/\lambda}$. This direct argument avoids any need to recover potentials by evaluating a density on potentially exceptional slices.

Further exercises**Exercise 8 (Unbalanced OT as constrained entropic problem).**

Let $C \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}_+^m$, $b \in \mathbb{R}_+^n$, $\tau > 0$. Define, for $r \in \mathbb{R}_+^m$, $a \in \mathbb{R}_+^m$,

$$\text{KL}(r \mid a) := \sum_{i=1}^m \left(r_i \log \frac{r_i}{a_i} - r_i + a_i \right),$$

and similarly for $\text{KL}(s \mid b)$. Consider the unbalanced OT problem

$$\min_{P \in \mathbb{R}_+^{m \times n}} \langle C, P \rangle + \tau \text{KL}(P \mathbf{1}_n \mid a) + \tau \text{KL}(P^\top \mathbf{1}_m \mid b). \quad (\text{UOT})$$

- (1) Show that (UOT) is equivalent to

$$\min_{\substack{P \geq 0 \\ P \mathbf{1}_n = r \\ P^\top \mathbf{1}_m = s}} \langle C, P \rangle + \tau \text{KL}(r \mid a) + \tau \text{KL}(s \mid b), \quad (\star)$$

and argue that if $(P^\star, r^\star, s^\star)$ solves $(\star)$, then $P^\star$ is an optimal *balanced* OT plan between the (relaxed) marginals $(r^\star, s^\star)$.

- (2) Write the Lagrangian of $(\star)$ using dual vectors $f \in \mathbb{R}^m$, $g \in \mathbb{R}^n$ for the constraints $P \mathbf{1}_n = r$ and $P^\top \mathbf{1}_m = s$.
- (3) Assuming we can swap $\inf_{P, r, s}$ and $\sup_{f, g}$, derive the dual involving only the two potentials f, g .
- (4) Assume a, b have equal total mass. Show that the dual objective tends pointwise to the balanced OT dual objective as $\tau \rightarrow +\infty$, and justify convergence of the optimal values.

Useful hints

- (1) Introduce the actual row and column sums as auxiliary variables.
- (2) Use $\langle f, r - P\mathbf{1} \rangle + \langle g, s - P^\top \mathbf{1} \rangle$.
- (3) Minimize first over P , then use the conjugate of KL.
- (4) Expand $e^{-f_i/\tau}$ for fixed potentials. For values, use a balanced competitor and coercivity of the KL penalties.

Detailed solution

(1) Introducing $r = P\mathbf{1}_n$ and $s = P^\top \mathbf{1}_m$ changes neither the objective nor the feasible plans. With r, s fixed, both penalties are constant, so $P^\star$ minimizes $\langle C, P \rangle$ over the transport polytope with marginals $(r^\star, s^\star)$.

(2) Choose the standard OT signs:

$$\mathcal{L} = \langle C, P \rangle + \tau \text{KL}(r \mid a) + \tau \text{KL}(s \mid b) + \langle f, r - P\mathbf{1}_n \rangle + \langle g, s - P^\top \mathbf{1}_m \rangle.$$

(3) The infimum in $P \geq 0$ is finite exactly when $f_i + g_j \leq C_{i,j}$, and then equals zero. For each row,

$$\inf_{r_i \geq 0} \{ \tau(r_i \log(r_i/a_i) - r_i + a_i) + f_i r_i \} = \tau a_i (1 - e^{-f_i/\tau}),$$

attained at $r_i = a_i e^{-f_i/\tau}$. Hence the dual is

$$\sup_{f_i + g_j \leq C_{i,j}} \left\{ \tau \sum_i a_i (1 - e^{-f_i/\tau}) + \tau \sum_j b_j (1 - e^{-g_j/\tau}) \right\}.$$

The finite-dimensional convex problem is feasible and coercive; positive a, b permit strict feasibility in its auxiliary-variable formulation, so strong duality holds. Notice the minus signs in the exponents: reversing the potential convention would also reverse the feasibility inequalities.

(4) For fixed f, g , $\tau(1 - e^{-f_i/\tau}) \rightarrow f_i$, giving the usual balanced dual objective $\langle a, f \rangle + \langle b, g \rangle$ with the same constraint.

For convergence of values, let M be the common input mass and let P_τ minimize UOT. A balanced optimal plan bounds its value above by the balanced OT value V . If $r = P_\tau \mathbf{1}, s = P_\tau^\top \mathbf{1}$ have common mass m , the log-sum inequality gives

$$\text{KL}(r \mid a) + \text{KL}(s \mid b) \geq 2(m \log(m/M) - m + M).$$

Together with $\langle C, P_\tau \rangle \geq -\|C\|_\infty m$, this bounds m uniformly for $\tau \geq 1$. The upper value bound then implies both KL penalties tend to zero. Any cluster point of P_τ has marginals a, b , so its cost is at least V . Nonnegativity of the penalties yields $\liminf V_\tau \geq V$, while the balanced competitor yields $V_\tau \leq V$. Thus $V_\tau \rightarrow V$. Equal mass is essential for this balanced limit.

Exercise 9 (Unbalanced OT + entropic regularization).

Let $C \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}_{++}^m$, $b \in \mathbb{R}_{++}^n$, and let $\tau > 0$, $\varepsilon > 0$. Denote

$$H(P) := \sum_{i,j} P_{i,j} (\log P_{i,j} - 1), \quad \text{KL}(r \mid a) := \sum_i \left(r_i \log \frac{r_i}{a_i} - r_i + a_i \right).$$

Consider

$$\min_{P \geq 0} \langle C, P \rangle + \varepsilon H(P) + \tau \text{KL}(P \mathbf{1} \mid a) + \tau \text{KL}(P^\top \mathbf{1} \mid b).$$

- (1) Compute the dual problem and show it involves only two dual potentials $f \in \mathbb{R}^m$, $g \in \mathbb{R}^n$.
- (2) For fixed g , give the maximizer in f (unbalanced Sinkhorn update).
- (3) Assume equal input masses and identify the pointwise limit of the dual objective as $\tau \rightarrow +\infty$.

Useful hints

- (1) Use the same multiplier signs as in Exercise 8; only the minimization in P changes.
- (2) Differentiate in f_i and isolate its exponential.
- (3) Keep f, g fixed and expand the marginal penalty terms.

Detailed solution

(1) Introducing r, s and adding $\langle f, r - P \mathbf{1} \rangle + \langle g, s - P^\top \mathbf{1} \rangle$ gives three separable minimizations. The row and column terms are as in Exercise 8, while

$$\inf_{P \geq 0} \left\{ \sum_{i,j} (C_{i,j} - f_i - g_j) P_{i,j} + \varepsilon H(P) \right\} = -\varepsilon \sum_{i,j} e^{(f_i + g_j - C_{i,j})/\varepsilon}.$$

Thus the dual is the unconstrained maximization of

$$\mathcal{D}(f, g) = \tau \sum_i a_i (1 - e^{-f_i/\tau}) + \tau \sum_j b_j (1 - e^{-g_j/\tau}) - \varepsilon \sum_{i,j} e^{(f_i + g_j - C_{i,j})/\varepsilon}.$$

The primal relation is $P_{i,j}^* = e^{(f_i^* + g_j^* - C_{i,j})/\varepsilon}$.

(2) Put $K_{i,j} = e^{-C_{i,j}/\varepsilon}$, $v_j = e^{g_j/\varepsilon}$, and $S_i = \sum_j K_{i,j} v_j > 0$. The stationarity equation is $a_i e^{-f_i/\tau} = e^{f_i/\varepsilon} S_i$. Hence

$$f_i^{\text{new}} = \frac{\varepsilon \tau}{\varepsilon + \tau} \log \frac{a_i}{S_i}, \quad u_i^{\text{new}} := e^{f_i^{\text{new}}/\varepsilon} = \left(\frac{a_i}{(Kv)_i} \right)^{\tau/(\tau + \varepsilon)}.$$

The scalar objective is strictly concave, so this critical point is its unique maximizer. The g update is analogous with $K^\top$.

(3) For fixed potentials, $\tau a_i (1 - e^{-f_i/\tau}) \rightarrow a_i f_i$. Therefore $\mathcal{D}(f, g)$ tends to $\langle a, f \rangle + \langle b, g \rangle - \varepsilon \sum_{i,j} e^{(f_i + g_j - C_{i,j})/\varepsilon}$, precisely the entropy-only balanced dual. The scaling exponent $\tau/(\tau + \varepsilon)$ tends to 1, recovering ordinary Sinkhorn updates.

Exercise 10 (φ -divergence, dual form and unbalanced OT).

Let α and β be finite nonnegative measures on a measurable space X , and let $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R} \cup \{+\infty\}$ be finite, convex and lower semicontinuous on $[0, \infty)$, with $\varphi(1) = 0$ and $\lim_{r \rightarrow \infty} \varphi(r)/r = +\infty$. The φ -divergence of α w.r.t. β is

$$D_\varphi(\alpha | \beta) := \begin{cases} \int \varphi\left(\frac{d\alpha}{d\beta}(x)\right) d\beta(x), & \alpha \ll \beta, \\ +\infty, & \text{otherwise.} \end{cases}$$

(1) Recall the convex conjugate $\varphi^*(t) := \sup_{u \geq 0} (ut - \varphi(u))$.

(2) Show that

$$D_\varphi(\alpha | \beta) = \sup_f \left\{ \int f d\alpha - \int \varphi^*(f) d\beta \right\}.$$

(3) Consider now the unbalanced OT problem

$$\min_{\pi \geq 0} \int c(x, y) d\pi(x, y) + D_\varphi(\pi_1 | \alpha) + D_\varphi(\pi_2 | \beta),$$

where π_1, π_2 are the first and second marginals of π . Using (2), derive a dual lower bound involving only two measurable test functions f, g . Show that equality holds whenever the indicated minimax interchange is justified; do not assert strong duality on an arbitrary measurable space without further hypotheses.

Useful hints

- (1) Extend φ by $+\infty$ to negative arguments.
- (2) Use Fenchel–Young and approximate the pointwise supremum using a countable dense family of slopes; on a singular set, take arbitrarily large test values.
- (3) Insert the two suprema and minimize the resulting linear functional over nonnegative measures. Keep track of the sign change to OT potentials.

Detailed solution

(1) After extending φ by $+\infty$ on $(-\infty, 0)$, its conjugate is $\varphi^*(t) = \sup_{u \geq 0} \{ut - \varphi(u)\}$. Superlinear growth makes $\varphi^*(t)$ finite for every finite t .

(2) Fenchel–Young gives $uf - \varphi^*(f) \leq \varphi(u)$, hence the required upper bound after integration. Conversely, enumerate the rational slopes and select, pointwise, the best of the first N affine minorants $ut - \varphi^*(t)$. The selector is measurable and takes finitely many values. These maxima increase to $\varphi(u)$; monotone convergence after subtracting one integrable affine minorant proves equality when $\alpha \ll \beta$.

If $\alpha \not\ll \beta$, there is a measurable set A with $\beta(A) = 0 < \alpha(A)$. Taking $f = M\mathbf{1}_A$ makes the variational value $M\alpha(A) - \varphi^*(0)\beta(X)$, which tends to $+\infty$. Thus the stated divergence formula is valid under the superlinear hypothesis. Without it, the usual divergence needs a finite recession-cost term for the singular part; the displayed $+\infty$ convention would not generally have this conjugate representation.

(3) Inserting the two variational formulas produces

$$\inf_{\pi \geq 0} \sup_{u, v} \left\{ \int (c + u + v) d\pi - \int \varphi^*(u) d\alpha - \int \varphi^*(v) d\beta \right\}.$$

Interchanging infimum and supremum gives a lower bound. The infimum in π is zero if $c(x, y) + u(x) + v(y) \geq 0$ everywhere, and $-\infty$ otherwise. In the standard OT convention $f = -u$, $g = -v$, the dual lower bound is

$$\sup_{f(x)+g(y) \leq c(x,y)} \left\{ - \int \varphi^*(-f) d\alpha - \int \varphi^*(-g) d\beta \right\}.$$

It equals the primal value if the minimax interchange is valid. For $\varphi(r) = \tau(r \log r - r + 1)$, its conjugate is $\tau(e^{t/\tau} - 1)$, recovering the signs in Exercise 8.

Exercise 11 (Barycentric projection from entropic OT).

Let $\alpha = \sum_{i=1}^m a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^n b_j \delta_{y_j}$ be probability measures with distinct support points and strictly positive weights, and let P_ε be the optimal plan of entropic OT between the weight vectors (a, b) for some finite cost $C_{i,j} = c(x_i, y_j)$. For x_i with $a_i > 0$ define the barycentric projection

$$T_\varepsilon(x_i) := \frac{1}{a_i} \sum_{j=1}^n (P_\varepsilon)_{i,j} y_j.$$

- (1) Suppose the optimal unregularized coupling is uniquely induced by a map $T : \{x_i\} \rightarrow \{y_j\}$. Denote the matrix of its graph plan $(\text{Id}, T)_\# \alpha$ by $P^\star$, and assume $P_\varepsilon \rightarrow P^\star$ entrywise as $\varepsilon \downarrow 0$. Show that

$$T_\varepsilon(x_i) \longrightarrow T(x_i) \quad \text{for every } i.$$

- (2) Assume now that $c(x, y) = \frac{1}{2} \|x - y\|^2$ and recall from Exercise 5 that

$$\nabla_{x_i} \mathcal{W}_\varepsilon(a, b) = a_i x_i - \sum_{j=1}^n (P_\varepsilon)_{i,j} y_j.$$

Express $T_\varepsilon(x_i)$ in terms of $\nabla_{x_i} \mathcal{W}_\varepsilon(a, b)$ and x_i , and explain why the displacement $T_\varepsilon(x_i) - x_i$, rather than the target point itself, is the negative cost gradient divided by a_i .

Useful hints

- (1) On a fixed finite support, weak convergence of plans gives convergence of every matrix entry.
 (2) Rearrange the gradient formula and distinguish a point from a displacement vector.

Detailed solution

- (1) Since $P_\varepsilon \rightarrow P^\star$ and $P^\star$ is supported on $\{(x_i, T(x_i))\}$, we have for each i

$$\sum_j (P_\varepsilon)_{i,j} \rightarrow \sum_j (P^\star)_{i,j} = a_i, \quad \sum_j (P_\varepsilon)_{i,j} y_j \rightarrow \sum_j (P^\star)_{i,j} y_j = a_i T(x_i).$$

Dividing by $a_i > 0$ gives $T_\varepsilon(x_i) \rightarrow T(x_i)$.

- (2) From the envelope-theorem computation we have

$$\nabla_{x_i} \mathcal{W}_\varepsilon(a, b) = a_i x_i - \sum_j (P_\varepsilon)_{i,j} y_j = a_i x_i - a_i T_\varepsilon(x_i),$$

hence

$$T_\varepsilon(x_i) = x_i - \frac{1}{a_i} \nabla_{x_i} \mathcal{W}_\varepsilon(a, b).$$

So the barycentric projection is obtained by moving x_i in the direction of *minus* the gradient of the entropic OT cost, with step size $1/a_i$. This makes explicit that T_ε is the OT “target point” encoded in the gradient of the cost.